

Given: Due:	Further Maths (Core Pure) HW 19				Name:	
Chapter 4: Roots or Polynomials						
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.						
91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)	
Useful resources: Your notes from recent lessons & your text book						
Questions - do these on lined paper						
1. The quadratic equation $2z^2 - 4z + 5 = 0$ has roots α and β. (i) Write down the values of $\alpha + \beta$ and $\alpha\beta$. [2] (ii) Find the quadratic equation with roots $3\alpha - 1, 3\beta - 1$. [3] (iii) Find the cubic equation which has roots α, β and $\alpha + \beta$. [4]						
2. The equation $z^3 + kz^2 - 4z - 12 = 0$ has roots α, β and γ. (i) Write down the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$, and express k in terms of α, β and γ. [3] (ii) For the case where $\gamma = -\alpha$, solve the equation and find the value of k. [4] (iii) For the case $k = 5$, find a cubic equation with roots $2 - \alpha, 2 - \beta, 2 - \gamma$. [4]						
3. The complex number $3 + 2i$ is a root of the equation $2x^3 + px^2 + 20x + q = 0$, where p and q are real numbers. (i) Find the other two roots of the cubic equation. [4] (ii) Find the values of p and q. [4]						
4. $f(z) = z^3 - 8z^2 + pz - 24$ where p is a real constant. Given that the equation $f(z) = 0$ has distinct roots $\alpha, \beta \text{ and } \left(\alpha + \frac{12}{\alpha} - \beta\right)$ (a) solve completely the equation $f(z) = 0$ (6) (b) Hence find the value of p. (2)						



5. The cubic equation $2z^3 + pz^2 + qz + r = 0$ has roots $\frac{\alpha}{k}, \alpha, k\alpha$.

(i) Express p, q and r in terms of k and α . [3]

(ii) Show that $2q^3 = p^3r$. [3]

(iii) Solve the equation for the case where $p = q = -3$. [4]

END OF HOMEWORK

Total = 46 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.

